



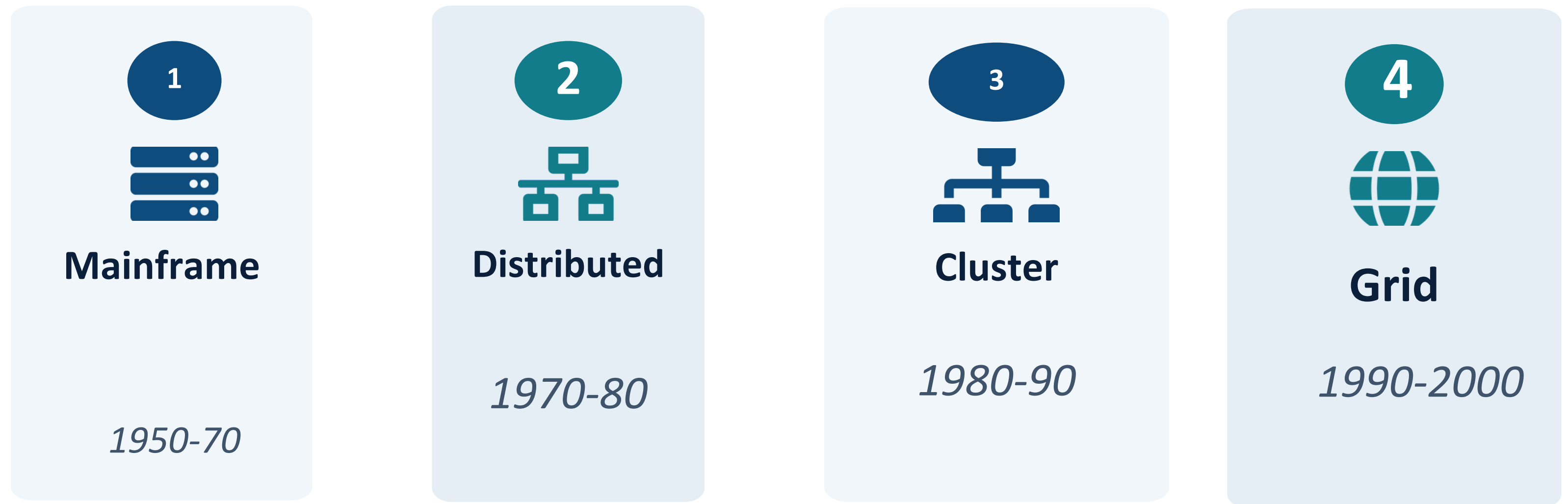
The

EVOLUTION OF CLOUD COMPUTING

From Mainframes (1950s) to the Modern Cloud — 8 Generations of Innovation

A 70-Year Journey

Each generation solved the previous one's biggest weakness — leading step by step to today's cloud.



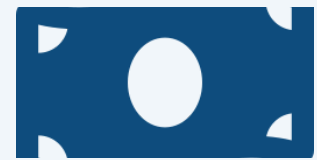
Cloud Computing

↑ *All eight streams converge into the cloud computing model we use today*

A 70-Year Journey

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5



Utility

late 90s-2000s

6



Virtualization

1980-present

7



Web 2.0

~2004

8



Service Orient.

2000s

Cloud Computing

↑ *All eight streams converge into the cloud computing model we use today*



1950 – 1970

Mainframes were the first true 'big iron' — highly powerful, reliable machines built to handle massive input-output operations with almost no downtime. They pushed computing power far beyond earlier systems, but their enormous cost meant only large institutions could afford them. This expense problem is exactly what pushed the industry toward cluster computing next.

Real-World Examples



Banking Systems

Overnight batch-processing
of millions of transactions



SABRE Airline System

One of the first mainframe-
powered reservation

systems



Government Census

Large-scale data processing
for national records



1970 – 1980

Distributed systems combined multiple independent computers that appeared to users as a single system. They introduced scalability, concurrency, continuous availability, and independence from failures — and didn't require every machine to sit in the same location, since they could communicate over networks. This model later evolved into cluster and grid computing.

Real-World Examples



ARPANET

Early academic network linking geographically separate computers



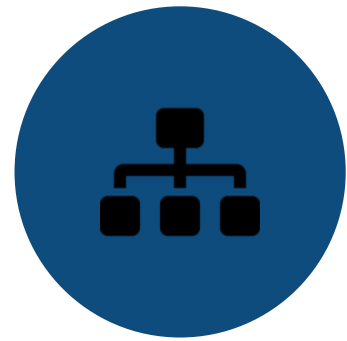
File-Sharing Networks

Multiple machines sharing resources as one logical system



Multi-Site Research Labs

Universities pooling computing power across campuses



Cluster Computing

1980 – 1990

Cluster computing emerged as a cheaper alternative to mainframes. Machines were linked by high-bandwidth networks, delivered comparable computing power, and new nodes could be added easily. This solved the cost problem — but the machines still needed to be geographically close, a limitation that grid computing would soon address.

Real-World Examples



Beowulf Clusters

Off-the-shelf PCs
networked for scientific



University Research Labs

Low-cost clusters replacing
expensive mainframes



Early Web Server Farms

Clustered machines sharing
website traffic loads



Grid Computing

1990 – 2000

Grid computing connected systems in entirely different geographic locations via the internet, forming a heterogeneous grid across multiple organizations. It solved the geography problem of clusters — but introduced new challenges around limited bandwidth over long distances. Grid computing is often called the direct predecessor of cloud computing.

Real-World Examples



SETI@home

Volunteers worldwide donated idle CPU time to analyze radio signals



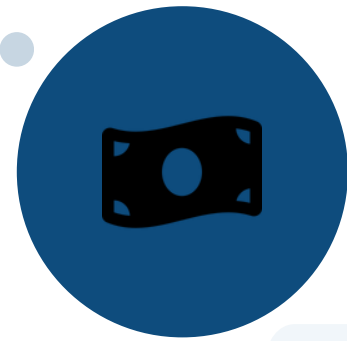
CERN LHC Computing Grid

Physics data processed across dozens of countries



Folding@home

Distributed protein-folding research across the globe



Utility Computing

Late 1990s – 2000s

Utility computing introduced a service-provisioning model where compute power, storage, and infrastructure are billed on a pay-per-use basis — just like electricity or water. This 'pay only for what you use' idea became one of the foundational principles of modern cloud pricing.

Real-World Examples



Sun Grid Utility

Early pay-per-CPU-hour
computing service



IBM On-Demand Computing

Metered enterprise
computing resources



Precursor to AWS

Pricing

Set the template for today's
pay-as-you-go cloud billing



Virtualization

1980 – Present

Virtualization creates a virtual layer over physical hardware, allowing multiple independent instances to run simultaneously on the same machine. Introduced decades ago, it remains the key technology underlying modern cloud platforms — hardware virtualization is still the most common form used today.

Real-World Examples



Amazon EC2

Thousands of virtual machines run on shared physical servers



VMware vCloud

Enterprise-grade hardware virtualization platform



VirtualBox / Hyper-V

Widely used desktop and enterprise virtualization tools



Gained Popularity ~2004

Web 2.0

Web 2.0 is the interface layer through which cloud services interact with everyday users — the shift from static pages to interactive, dynamic web experiences. It's the technology that made modern social media possible and greatly increased flexibility across the web.

Real-World Examples



Google Maps

Interactive, dynamic mapping
in the browser



Facebook

Social media built entirely on
interactive web pages



Twitter

Real-time, dynamic content
sharing at global scale



Service Orientation

2000s

Service orientation became the reference model for cloud computing, supporting low-cost, flexible, and evolvable applications. It introduced two ideas still central to the cloud today: Quality of Service (QoS) with Service Level Agreements (SLAs), and Software as a Service (SaaS).

Real-World Examples



Salesforce

One of the first major SaaS companies, with contractual



SLA-Backed Hosting

Providers guaranteeing uptime and performance



Early Web Services (SOAP/XML)

Standardized, service-oriented application architecture

All Roads Lead to the Cloud

Cloud Computing means storing and accessing data and programs on **remote servers hosted on the internet**, instead of a computer's hard drive or a local server.



**Mainframe
Power**



**Distributed
Access**



**Virtualized
Resources**



**Pay-Per-Use
Billing**



**SaaS &
SLAs**

Today's cloud platforms (AWS, Azure, Google Cloud) are the sum of every generation before them.



PUBLIC vs PRIVATE vs HYBRID CLOUD

Choosing the Right Cloud Deployment Model

Public

Private

Hybrid

Three Deployment Models, One Decision

Who owns the infrastructure — and who else shares it with you.



Public Cloud

Shared infrastructure owned by a third-party provider. Multi-tenant, pay-as-you-go.

Multi-Tenant



Private Cloud

Dedicated infrastructure for one organization. Single-tenant, maximum control.

Single-Tenant



Hybrid Cloud

A strategic mix of both — sensitive workloads stay private, rest goes public.

Mixed

PUBLIC CLOUD



Owned & Shared by a Provider

Infrastructure and services fully owned, operated, and maintained by a third-party provider, delivered over the public internet and shared among multiple organizations — a setup known as multi-tenancy.

Core Concept: *You pay only for the exact resources you consume.*

Well-Known Examples

 **Amazon Web
Services (AWS)**

 **Microsoft
Azure**

 **Google Cloud
Platform (GCP)**

Public Cloud — Advantages & Disadvantages

Advantages

- ✓ Cost-efficient — no hardware to buy, pure pay-as-you-go
- ✓ Automatic updates, patches & maintenance by the provider
- ✓ High accessibility — usable from anywhere with internet

Disadvantages

- ⚠ Security & privacy concerns on shared servers
- ⚠ Limited control over underlying infrastructure
- ⚠ Reliance on internet connectivity for performance
- ⚠ May not meet strict compliance needs (e.g. HIPAA)
- ⚠ Unmonitored usage can cause cost overruns

PRIVATE CLOUD



Dedicated to One Organization

A computing environment dedicated entirely to a single organization. It can be hosted on-site or by a third party, but resources are never shared with other tenants.

Core Concept: *Maximum control and isolation for sensitive operations.*

Well-Known Examples



 HPE / Dell
Infrastructure

 Bespoke Internal
Data Centers

Advantages

- ✓ Higher security — full control to customize protections
- ✓ Deep customization of infrastructure and services
- ✓ Greater privacy over who accesses data and resources

Disadvantages

- ⚠ Higher cost — dedicated hardware & networking are expensive
- ⚠ Limited scalability, capped by physical hardware
- ⚠ Technical complexity requires specialized IT expertise
- ⚠ Still vulnerable to breaches and cyberattacks
- ⚠ Lack of standardization complicates future migration
- ⚠ Ongoing maintenance and upgrades take real resources

HYBRID CLOUD



The Best of Both Worlds

Combines public and private cloud environments, letting data and applications move between them — sensitive, mission-critical workloads run on the private cloud, while the public cloud absorbs burst traffic or less-sensitive workloads.

Core Concept: *Optimizing for security, performance, and cost simultaneously.*

Well-Known Examples

 **AWS
Outposts**

 **Azure
Stack**

 **IBM Hybrid
Cloud**

Hybrid Cloud — Advantages & Disadvantages

Advantages

- ✓ Flexibility — sensitive data stays private, public adds reach
- ✓ Scalability — workloads shift between clouds on demand
- ✓ Balanced security with public cloud's cost savings

Disadvantages

- ⚠ Complexity of integrating two different environments
- ⚠ Often costlier overall than one model alone
- ⚠ Security risks at the seams between environments
- ⚠ Data governance challenges across environments (GDPR, HIPAA)
- ⚠ Network latency between environments
- ⚠ Integration & compatibility challenges
- ⚠ Possible vendor lock-in across multiple providers

Head-to-Head Comparison

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Resource Allocation	Shared / multi-tenant	Dedicated / single-tenant	Mix of shared & dedicated
Data Storage	Provider's remote servers	Internal, isolated servers	Sensitive data private, rest public
Pricing Model	Pay-as-you-go, variable	High fixed upfront cost	Mixed: fixed + variable
Management	Fully by provider	Internal IT team	Co-managed
Scalability	Near-infinite, instant	Limited by hardware	Flexible — bursts to public
Overall Cost	Cost-effective at scale	Expensive to build/maintain	Optimized by workload routing

Which One Should You Choose?



Choose Public

Startups, variable workloads,
and teams that want zero
hardware overhead.



Choose Private

Regulated industries, sensitive
data, and orgs needing full
infrastructure control.



Choose Hybrid

Businesses balancing sensitive
workloads with the need to
scale on demand.

There's no universally 'best' model — the right choice depends on your workload sensitivity, budget, and need for scale.

Thank You!